

Multiple Choice (10 marks)

1. What is meant by the term muscle fatigue?
 - (a) Muscle fatigue is the result of a decrease in muscle temperature during exercise.
 - (b) Muscle fatigue is due to lack of physical exercise.
 - (c) Muscle fatigue is a result of excessive calcium buildup in the muscle tissue.
 - (d) Muscle fatigue is due to overhydration.
 - (e) Muscle fatigue is due to a permanent loss of muscle fibers.
2. What are the chief differences between plant and animal cells?
 - (a) Animal cells have larger vacuoles than plant cells
 - (b) Animal cells have a rigid cellulose wall
 - (c) Animal cells contain chloroplasts
 - (d) Plant cells are typically smaller than animal cells
 - (e) Animal cells can perform photosynthesis, while plant cells cannot
3. All of the following may serve as intracellular messengers EXCEPT
 - (a) calcium ions
 - (b) cAMP
 - (c) acetylcholine
 - (d) inositol 1,4,5-triphosphate
4. If a couple has had three sons and the woman is pregnant with their fourth child, what is the probability that child 4 will also be male?
 - (a) $3/4$
 - (b) $2/3$
 - (c) $1/16$
 - (d) $1/4$
 - (e) $1/5$
5. Cancer cells grown in culture are similar to normal cells grown in culture in that they
 - (a) do not have the ability to metastasize
 - (b) divide an indefinite number of times
 - (c) undergo programmed cell death
 - (d) can grow without a nutrient supply
 - (e) do not require oxygen to survive

True / False (10 marks)

Answer True or False

6. True or False: Some aspects of social exclusion: do they influence suicide mortality?
7. True or False: Aripiprazole: a new risk factor for pathological gambling?
8. True or False: Cold knife conization vs. LEEP. Are they the same procedure?
9. True or False: Does the use of atypical antipsychotics as adjunctive therapy in depression result in cost savings?
10. True or False: Substance use and HIV-related sexual behaviors among US high school students: are they related?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the behavioral differences between kinesis and taxis in animals, providing examples of each and analyzing their ecological significance.
12. Discuss the differences between fork ferns and true ferns, focusing on their morphological features and ecological adaptations, and analyze the significance of vascular tissues in *Psilotum*.

End of Paper